TailDaam

Dataset Proposal: Pakistan fuel prices and Brent risk

HACK INSIGHT 2026 | Track: Energy demand | Danyal Abbas

1. Question and why this data

Pakistan's petrol and high-speed diesel prices are administered at fortnightly reviews, but public argument about them is rarely tied to a traceable record. TailDaam answers two narrow questions: what was the notified domestic price, and how volatile is the global Brent benchmark that feeds Pakistan's import bill? The primary model targets next-month Brent realised volatility, not the administered domestic level. Domestic prices are used for a descriptive pass-through calculation and for the cited insight agent.

2. Sources, volume and licence

Source	Files	Clean rows	Role	Licence
PBS weekly SPI	41	82	Domestic petrol/HSD notified prices	Government open publication
PBS monthly annex	1	36	17 cities + national average	Government open publication
US EIA	11	84,546	Daily spot series; Brent is the model target	Public domain (US Gov)
World Bank Pink Sheet	1	6,720	Monthly energy benchmark / cross-check	CC BY 4.0
Total: 54 raw files | 91,384 clean rows | 3 publishers				

3. What is used for training, and what is not

The primary model is a HAR-style ridge regression trained on the EIA daily Brent Europe spot series (target: next calendar month annualised realised volatility). The features are lagged monthly, quarterly and annual realised volatility. No PBS or World Bank observation is used to fit that model. PBS weekly data is used for the descriptive Pakistan pass-through calculation (39 overlapping weeks; reported as descriptive, not causal). PBS city data, EIA and World Bank rows are indexed for the RAG agent so answers can quote them with row IDs. The World Bank series is an independent benchmark, not a training input.

4. Collection, provenance and independence

All 54 raw files were fetched from the original publishers on 20-21 October 2026, during the event. No data is taken from a Kaggle, GitHub or blog re-upload. data/manifest.csv records the source URL, query, fetch time, SHA-256 and licence for every file. The three publishers are independent: PBS (Pakistan), EIA (United States) and the World Bank. The volume is 91,384 clean rows, above the 50,000-row minimum, and the sources include material outside the suggested Energy-demand track list.

5. Quality, limitations and honesty controls

The deterministic quality agent applies 14 rules and logs every FIX, FLAG and FAIL decision. The current status is WARN: 1 fix and 4 flags, including the 9 April 2026 HSD outlier (522.29 PKR/L against 380.56 petrol), which is kept and annotated rather than silently removed. No synthetic, interpolated or AI-generated observations are used for training or evaluation. The PBS weekly history is short (41 reports) with genuine publication gaps; the monthly city annex covers September 2026; domestic prices are administered. These limits are stated in the README and Summary.pdf.

6. Reproducibility

Every layer is rebuilt with one command: `make data` runs acquire -> clean -> quality -> gold. `make model` refits the declared baseline and the pre-registered challenger; `make test` runs the contract and honesty tests; `make all` runs the full chain and regenerates Summary.pdf. The fitted model, gold tables, citation registry and all metrics are committed. The Hugging Face Space is a static dashboard over the verified API; the API health endpoint reports the embedding mode and indexed fact count.